

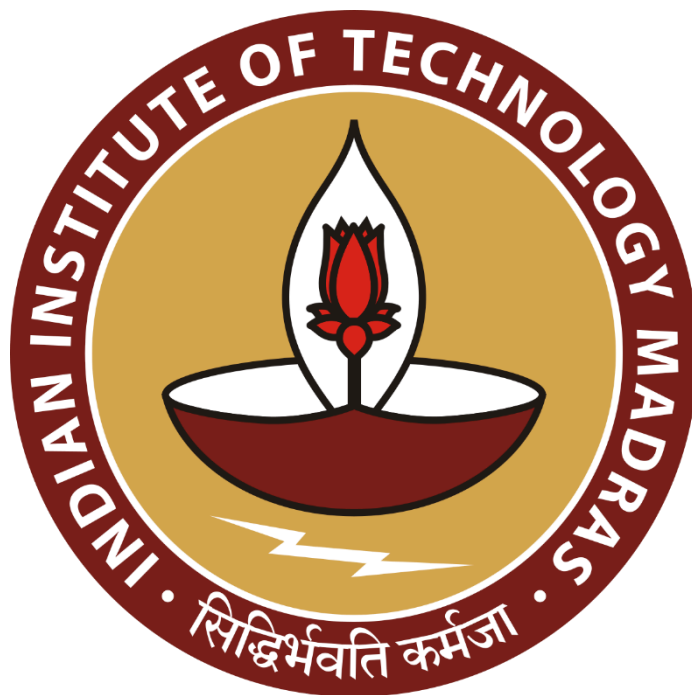
# **Inventory Management and Purchase Optimization Analysis for Rajesh Surgical**

**A Proposal report for the BDM capstone Project**

Submitted by

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## **Declaration Statement**

I am working on a Project Title "**Inventory Management and Purchase Optimization Analysis for Rajesh Surgical.**" I extend my appreciation to **Rajesh Surgical**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.



Signature of Candidate: **(Digital Signature)**

Name: Pranjal Dubey

Date: 11-Sep-2025

# 1 Executive Summary

The project title is "Inventory Management and Purchase Optimization Analysis for Rajesh Surgical." This report outlines a data-driven initiative to improve the operational efficiency and profitability of a wholesale surgical supply business by leveraging its historical purchase data. The project will use Microsoft Excel and Python to analyze data and provide a solution to a business problem.

Rajesh Surgical is a medical supplies wholesaler located in Moradabad. The company serves as a crucial link between manufacturers and its diverse customer base of hospitals, clinics, and retail stores, providing a wide range of surgical items.

The main business problem is the company's purchasing and inventory approach. This has resulted in inefficiencies such as overstocking of certain items, which ties up capital, and understocking of high-demand products, which can lead to lost revenue.

To solve this problem, the project will analyze three years of purchase data to identify key trends and seasonal patterns. By applying time-series analysis, future purchase needs will be forecasted. The expected outcome is a set of actionable recommendations to optimize inventory levels and improve the company's purchasing strategy.

# 2 Organization Background

Rajesh Surgical is a medical wholesale business located in Kaji Ki Sarai, Kothiwalan, Moradabad, Uttar Pradesh. The business was founded by Mr. Rajesh Sharma and has been operational for 30 years. For the past 9-10 years, it has been managed by his son, Ankit Pandey, and his wife, Ishita. With a substantial turnover in crores (more than 9 crores), the company is a significant player in the regional medical/surgical supplies market.

The business acts as a vital link in the medical supply chain, sourcing surgical products directly from various manufacturers. While the business has a strong operational foundation, its purchasing and inventory decisions have historically been made based on past experience. Data is recorded digitally, but the focus has been on record-keeping rather than comprehensive data analysis, which this project aims to address.

### 3 Problem Statement

The business is facing challenges in its current inventory and purchasing practices due to a lack of data-driven insights. The absence of a systematic approach to analyzing past purchases has led to inefficiencies, including:

- 3.1 **Problem Statement 1: Ineffective Inventory Planning & Suboptimal Purchase Strategy.**  
Without a clear understanding of purchasing patterns and supply needs, Rajesh Surgical struggles to maintain optimal stock levels. This results in an excess of slow-moving items, which ties up capital, or insufficient stock of high-demand products, leading to lost sales and potential customer dissatisfaction. Furthermore, the company's current purchasing methods are not based on a comprehensive analysis of historical purchase data. This makes it difficult to predict which items should be purchased, in what quantity, and when.
- 3.2 **Problem statement 2: Inefficient Cost Management-** The company's current purchasing methods do not account for price fluctuations or potential cost-saving opportunities. A data-driven approach is needed to identify items with variable pricing and recommend optimal times for bulk purchases to reduce overall acquisition costs.

### 4 Background of the Problem

#### **Problem 1: Ineffective Inventory Planning & Suboptimal Purchase Strategy**

The business has been operating with a transactional data management system that records purchases without providing the necessary analytical tools to derive meaningful insights. As a result, inventory and purchasing decisions are often made based on anecdotal evidence or historical precedent rather than on a precise understanding of purchase trends.

This lack of a data-driven approach is a recurring problem that directly affects the business's profitability. A large number of items in stock lead to capital being tied up and could expire, while shortages of key items can result in lost revenue and harm the company's reputation.

#### **Problem 2: Inefficient Cost Management**

The company's current purchasing methods do not account for price fluctuations or potential cost-saving opportunities. A data-driven approach is needed to identify items with variable pricing and recommend optimal times for bulk purchases to reduce overall acquisition costs. This project is a direct effort to address this knowledge gap by converting raw transactional data into a valuable asset for strategic decision-making.

## 5 Problem Solving Approach

The project will employ a systematic, data-driven methodology to address the identified business problems. The approach will follow these key stages:

### Data Collection-

The project will utilize a primary dataset of historical purchase records provided by Rajesh Surgical, covering the period from March 2022 to April 2025. This data includes key variables such as Date of Purchase, Item Name, Quantity, and Unit Price.

### Methods-

The project will employ a systematic, data-driven methodology to address the identified business problems. The approach will follow these key stages:

- **Data Cleaning:** The initial phase will involve data cleaning, which is crucial for ensuring the accuracy and reliability of the analysis. This process will involve separating concatenated columns, handling non-uniform product names (e.g., spelling differences or abbreviations), and addressing any missing values or inconsistencies to ensure the data is in a usable format.
- **Exploratory Data Analysis (EDA):** Visual and statistical methods will be used to explore the cleaned data. This will include creating bar charts to show total purchasing volume by item, line graphs to visualize trends in demand over time, and pie charts to display the percentage of total purchases for each product category. This EDA will help uncover preliminary insights into which items are in high demand and whether there are specific times of the year when purchasing activity peaks.
- **Time-Series Analysis:** The core of the analysis will involve applying time-series analysis techniques to the purchase data. This method will be used to forecast future purchase requirements for different surgical items. The analysis will identify trends and seasonal components in the data.

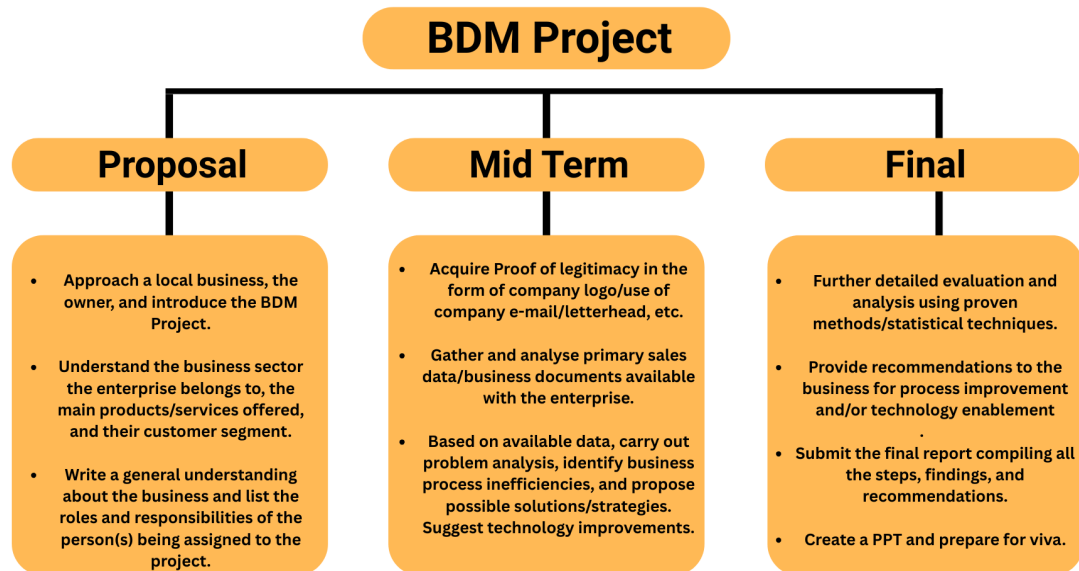
### Tools Used-

The project will primarily use Microsoft Excel and Python for data processing, analysis, and visualization.

- **Python:** The programming language will be used to perform data cleaning like handling nulls, proper formatting of columns, separating concatenated columns, handling non-uniform product names (e.g., spelling differences or abbreviations), and addressing any missing values or inconsistencies to ensure the data is in a usable format. Libraries like Pandas will be utilized to handle and manipulate the data, while libraries like Matplotlib or Seaborn will be used for creating visualisations.
- **Excel:** Will be used for data processing, including data filtering, sorting, and creating pivot tables for preliminary analysis. It will also be used to generate visualizations like bar charts and line graphs.

## 6 Expected Timeline

### 5.1 Work Breakdown Structure:



### 5.2 Gantt chart

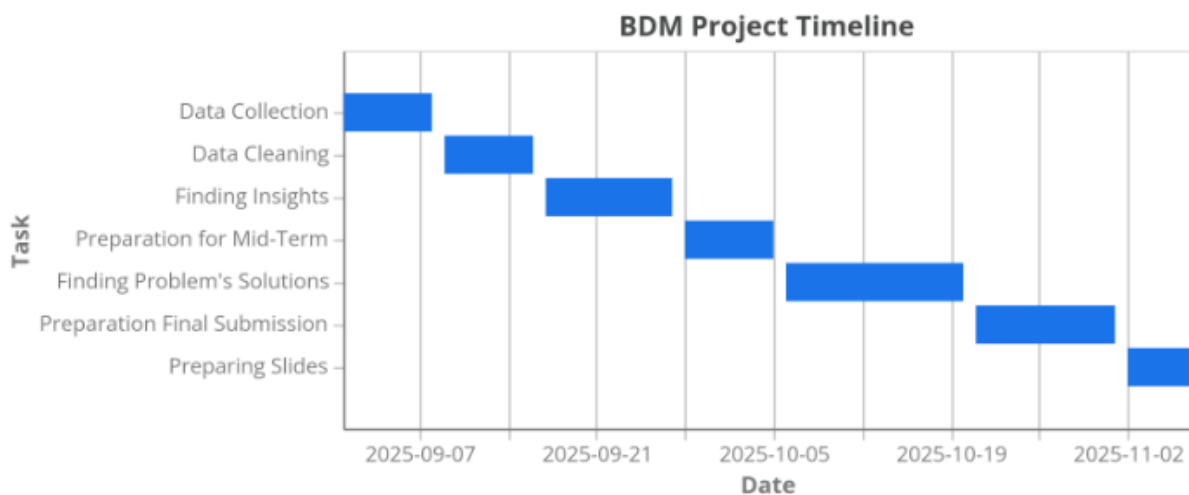


Figure 1 Expected timeline for completion of project.

## 6 Expected Outcome

The project's outcomes will directly address the business problems by providing clear, data-driven insights and recommendations to Rajesh Surgical. The key deliverables are expected to include:

- **Optimized Inventory Levels:** By forecasting future purchase needs, the company can avoid both overstocking and understocking, ensuring that capital is not unnecessarily tied up in inventory.
- **Cost Reduction:** The analysis will identify opportunities to reduce purchasing costs by highlighting trends and potential areas for negotiation with manufacturers.
- **Improved Decision-Making:** The findings will empower the management of Rajesh Surgical to make more informed, strategic decisions regarding inventory, purchasing, and supply chain management.
- **Actionable Insights:** The report will provide specific, easy-to-understand insights, such as which surgical items have the highest and lowest purchasing frequency, and how purchasing trends are affected by seasonality.